51 Key Concepts in Economics

Simple Explanations for Exam Preparation

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Quick-revision reference · one concept at a time

Theories, Models & Theorems — At a Glance

1 Pareto Efficiency

A state where no one can be made better off without making someone else worse off. It is the benchmark for economic efficiency in resource allocation. It says nothing about fairness — a hugely unequal outcome can still be Pareto efficient as long as no costless improvement remains for anyone.

2 Coase Theorem

If property rights are clearly defined and bargaining costs are zero, private parties can negotiate their own solution to externalities (like pollution) and reach an efficient outcome — regardless of who holds the rights. In the real world, high transaction costs often block this, justifying government intervention.

3 Arrow Impossibility Theorem

Kenneth Arrow proved that no voting system can convert individual preferences into a consistent collective ranking while satisfying a few reasonable fairness conditions at once. In short, there is no perfect democratic method for aggregating everyone's choices into one rational social preference order.

4 Nash Equilibrium

A situation in a game where each player's strategy is the best response to others' strategies, so no one gains by changing alone. Everyone is doing the best they can given what rivals do. It is the central solution concept in game theory.

5 Heckscher–Ohlin Theory

A country exports goods that use its abundant, cheap factor intensively and imports goods needing its scarce factor. So a labour-rich country exports labour-intensive goods. Trade patterns are explained by differences in countries' factor endowments rather than technology.

6 Ricardian Trade Theory

David Ricardo showed that countries gain from trade by specializing in goods where they have a comparative (relative) advantage, even if one country is better at producing everything. Differences in labour productivity drive mutually beneficial trade.

7 Comparative Advantage

A country (or person) should specialize in producing what it can make at the lowest opportunity cost, then trade. Even an all-round less efficient producer benefits by focusing on its relatively best activity. This is the core logic behind gains from trade.

8 Stolper-Samuelson Theorem

A rise in the price of a good (e.g. from a tariff) raises the real return to the factor used intensively in making it, and lowers the other factor's return. So protection helps a country's scarce factor and hurts its abundant factor.

9 Rybczynski Theorem

At constant prices, an increase in the supply of one factor (say labour) raises output of the good using that factor intensively and reduces output of the other good. It links changes in factor endowments to the composition of production.

10 Walras' Law

In a general equilibrium with many markets, if all markets but one are in balance, the last one must also balance. The total value of excess demand across all markets sums to zero, so markets are interconnected and clear together.

11 Fisher Equation

The nominal interest rate roughly equals the real interest rate plus expected inflation. It separates the actual money return from the inflation-adjusted return, showing how lenders price in expected price rises to protect purchasing power.

12 Quantity Theory of Money

MV = PT. The money supply times its velocity equals price level times transactions. If velocity and output are stable, increasing the money supply mainly raises prices. It is the classical foundation for linking money growth to inflation.

13 IS-LM Model

A framework showing how the goods market (IS curve) and money market (LM curve) jointly determine interest rates and output. Their intersection gives equilibrium. It illustrates how fiscal and monetary policy shift the economy in the short run.

14 AD-AS Model

Aggregate Demand and Aggregate Supply curves together determine the economy's overall price level and output. Shifts in either curve explain inflation, recession, and growth, making it a basic tool for analyzing macroeconomic fluctuations and policy.

15 Solow Growth Model

Long-run growth comes from capital accumulation, labour, and especially technological progress. Because capital has diminishing returns, economies reach a steady state; sustained growth in living standards ultimately depends on technology, not just saving more.

16 Harrod-Domar Model

Economic growth depends on the savings rate and the capital-output ratio: more saving and investment means faster growth. It was an early growth model stressing capital, but its 'knife-edge' instability made steady growth seem fragile.

17 Ramsey Model

A growth model where households choose saving and consumption over time to maximize lifetime welfare. Unlike Solow's fixed saving rate, the saving rate is derived from optimal forward-looking behaviour, forming the basis of modern growth theory.

18 Phillips Curve

An inverse short-run relationship between unemployment and inflation: lower unemployment tends to come with higher inflation. Later economists added expectations, arguing the trade-off vanishes in the long run, where unemployment returns to its natural rate.

19 Kuznets Curve

Inequality first rises then falls as a country develops, tracing an inverted-U. Early industrialization widens income gaps, but later growth and policy narrow them. An environmental version applies the same shape to pollution and income.

20 Laffer Curve

Tax revenue and tax rates have a hump-shaped relationship. At 0% and 100% rates revenue is zero; somewhere between lies a revenue-maximizing rate. Beyond that point, raising rates discourages activity so much that revenue actually falls.

21 CAPM

The Capital Asset Pricing Model prices an asset's expected return as the risk-free rate plus a premium for its market risk (beta). Only non-diversifiable risk is rewarded. It links expected return to systematic risk in financial markets.

22 Black-Scholes Model

A formula for pricing financial options (the right to buy or sell an asset at a set price). It uses factors like volatility, time, and interest rates. It revolutionized derivatives markets by giving traders a rigorous valuation method.

23 Modigliani-Miller Theorem

Under ideal conditions (no taxes, no bankruptcy costs), a firm's value does not depend on whether it finances itself with debt or equity. Capital structure is irrelevant; value comes from the firm's assets, not its financing mix.

24 Slutsky Equation

Breaks down how a price change affects demand into two parts: the substitution effect (switching toward cheaper goods) and the income effect (changed real purchasing power). It is fundamental to consumer demand theory.

25 Cobb-Douglas Function

A widely used production (or utility) function, Q = A·L^a·K^b, where output depends on labour and capital with constant factor shares. Its convenient properties make it a workhorse in growth, production, and distribution analysis.

26 CES Function

The Constant Elasticity of Substitution function generalizes production/utility functions by allowing a fixed, adjustable ease of substituting one input for another. Cobb-Douglas and fixed-proportion functions are special cases of it.

27 Arrow-Debreu Equilibrium

A rigorous proof that under certain conditions, competitive markets with prices for every good (across time and states of the world) can reach a general equilibrium where supply equals demand everywhere. It is the formal backbone of general equilibrium theory.

28 Cournot Model

A model of oligopoly where firms compete by choosing quantities to produce simultaneously, each assuming rivals' output is fixed. The result lies between monopoly and perfect competition, with price above the competitive level.

29 Bertrand Model

An oligopoly model where firms compete on price rather than quantity. With identical products, price competition drives prices down to marginal cost — the competitive outcome — even with just two firms, the 'Bertrand paradox'.

30 Stackelberg Model

An oligopoly model with a leader firm that chooses output first and follower firms that react afterward. The first-mover leader gains an advantage by committing to a larger output, earning higher profit than the followers.

31 Mundell-Fleming Model

Extends IS-LM to open economies, adding exchange rates and capital flows. It shows fiscal and monetary policy work very differently under fixed versus floating exchange rates, and you cannot have free capital flow, fixed rates, and independent monetary policy at once.

32 Tinbergen Theorem

To achieve a given number of independent policy targets, policymakers need at least an equal number of independent policy instruments. You cannot reliably hit two goals with only one tool.

33 Median Voter Theorem

Under majority rule with single-peaked preferences, the outcome reflects the preference of the median voter. This is why competing parties often converge toward the political center to capture the decisive middle voter.

34 Tiebout Model

People 'vote with their feet' by moving to the local jurisdiction whose mix of taxes and public services best suits them. This competition among local governments can efficiently reveal preferences and supply public goods.

35 Lindahl Equilibrium

A solution for funding public goods where each person pays a tax price equal to their personal marginal benefit. Everyone agreeing on the quantity yields an efficient level of the public good — though preferences are hard to reveal honestly.

36 Lewis Model

A development model with a 'dual economy': surplus labour from a low-productivity traditional (agricultural) sector shifts to the modern industrial sector, fuelling growth and capital accumulation until the surplus labour is exhausted.

37 Harris-Todaro Model

Explains rural-to-urban migration: people migrate based on expected urban wages (actual wage times the chance of getting a job), not just the wage itself. This can produce persistent urban unemployment even with migration.

38 Principal-Agent Model

Analyzes problems when one party (agent) acts on behalf of another (principal) but has different interests and hidden information or actions — like employers and employees. It studies how to design contracts and incentives to align their goals.

39 Signaling Model

When one side has more information, it can credibly 'signal' quality through costly actions — e.g. education signaling worker ability to employers. The signal works only if it is too costly for low-quality types to fake.

40 Efficient Market Hypothesis

Asset prices fully reflect all available information, so it is impossible to consistently beat the market using known information. Prices move only on new, unpredictable news, implying a near-random walk in stock prices.

41 Gauss-Markov Theorem

In a linear regression meeting standard assumptions, the Ordinary Least Squares (OLS) estimator is the Best Linear Unbiased Estimator (BLUE) — it has the smallest variance among all unbiased linear estimators. It justifies OLS in econometrics.

42 Fundamental Welfare Theorems

First: every competitive market equilibrium is Pareto efficient. Second: any efficient allocation can be achieved as a market equilibrium with the right initial redistribution. Together they formalize Adam Smith's 'invisible hand'.

43 Marshall-Lerner Condition

A currency devaluation improves a country's trade balance only if the sum of the price elasticities of export and import demand exceeds one. If demand is too inelastic, devaluation can worsen the trade balance.

44 Purchasing Power Parity

Exchange rates should adjust so that a basket of goods costs the same across countries when priced in a common currency. In the long run, currencies move toward equalizing purchasing power; the 'Big Mac Index' is a popular illustration.

45 Interest Rate Parity

Differences in interest rates between two countries are offset by expected changes in their exchange rate. Otherwise, investors could profit risk-free through arbitrage, so currency forward premiums reflect interest rate gaps.

46 Keynesian Multiplier

An initial rise in spending (e.g. government investment) leads to a larger total increase in national income, because each recipient re-spends part of the money. The size depends on how much people spend versus save out of extra income.

47 Accelerator Principle

Investment depends on the rate of change of output, not its level. When demand grows, firms invest heavily to expand capacity; when growth slows, investment can fall sharply — amplifying business cycle swings.

48 Golden Rule of Growth

The savings rate that maximizes long-run consumption per person in the Solow model. Saving too little or too much both reduce sustainable consumption; the golden rule finds the optimal capital stock for the highest steady-state consumption.

49 Lucas Critique

You cannot reliably predict policy effects using past statistical relationships, because people's expectations and behaviour change when policy changes. Models must be built on stable, deep parameters (preferences, technology), not historical correlations.

50 Rational Expectations Hypothesis

People form expectations using all available information and do not make systematic errors, effectively anticipating policy on average. This implies predictable government actions have limited real effects, since people adjust in advance.